

When Does Fusing Hand-Crafted Knowledge With Learned Representations Pay?

(a) Controlled Instrument

Frozen Recipe: SGD 0.1, Cosine, 200 Epochs, Batch 128, Crop and Flip

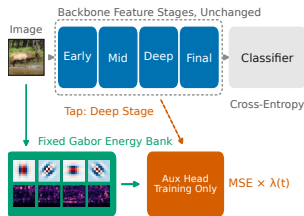
Committed Subset Indices, Per Dataset and Fraction

Pinned Filter Bank (Fingerprinted)

Every Intervention Sees
Byte-Identical Images

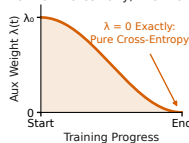
Deviations Quarantined,
Never Headline

(b) MomentAux: The Fused Prior

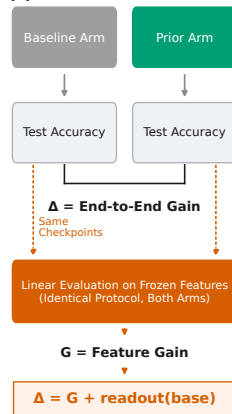


+2% Training Compute, +0 at Inference

Prior Dominates Early, Then Vanishes



(c) Two Measurements



(d) Comparators, By Cost

ImageNet Transfer
External Data

SimCLR, SimSiam, DINO
2× Compute

FitNets Learned Teacher
2× Compute

DeiT Augmentation
Recipe Change

HOG Target
1.02×

MomentAux (Ours)
1.02×

All Fused Into the Same Frozen Recipe;
Combinations Measured Too

(e) When Fusion Pays

STACK

Prior + Augmentation
Food-101 5%



SUBSTITUTE

Prior + SimCLR Init
ViT-Tiny CIFAR-100 10%



INTERFERE

Prior + ImageNet Init
CIFAR-100 7%

